

Group 4

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1 Verification and validation are the processes in which we check a product against its specifications and the expectations of the users who will be using it.

Verification:

It is the process of checking whether the product is being built correctly according to system specifications.

“Are we building the product right?”

Validation:

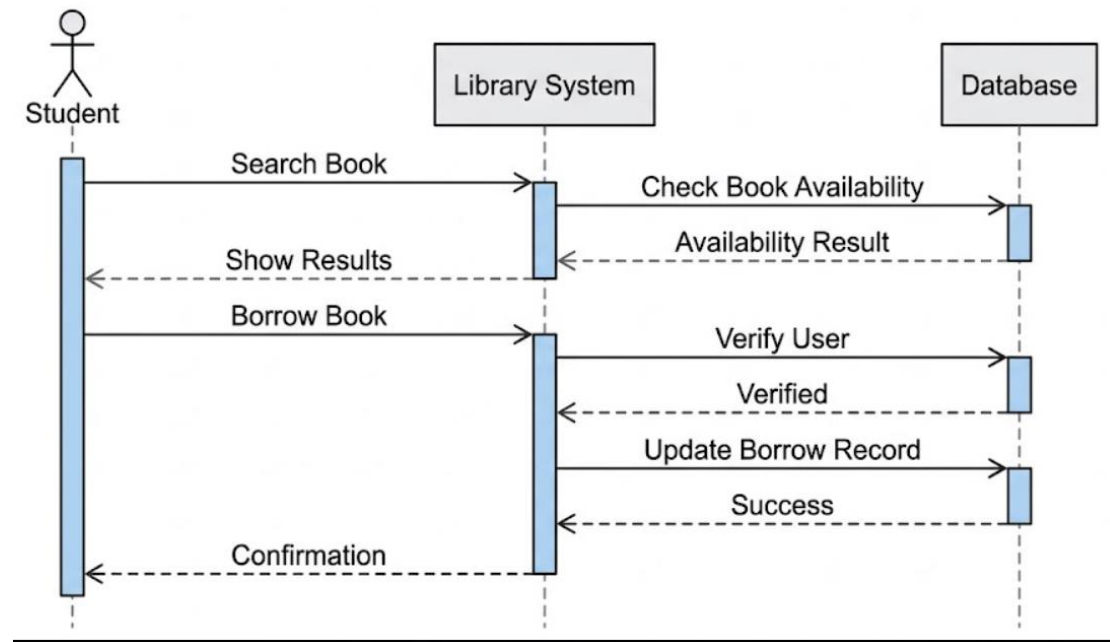
It is the process of checking whether the product meets the user needs and expectations.

“Are we building the right product?”

Role in software development:

Verification and validation ensure that the software meets system specifications and user requirements before deployment.

2. Sequence Diagram – Library Book Borrowing System

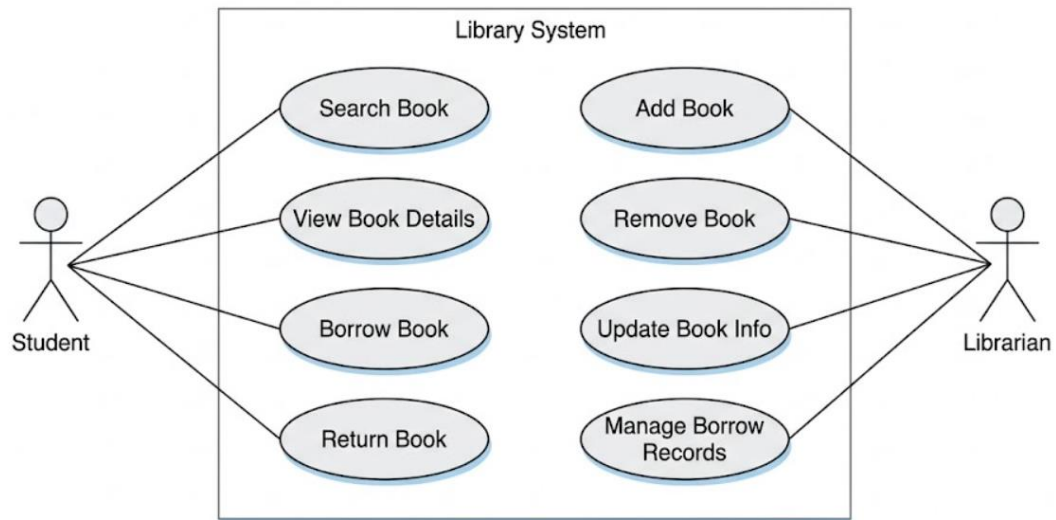


This diagram shows **how the system works step by step over time** when a student borrows a book.

Flow:

1. The **student searches for a book**.
2. The **library system checks availability** in the database.
3. The **database returns the result** (available or not).
4. The system **shows results to the student**.
5. The student chooses to **borrow the book**.
6. The system **verifies the user** in the database.
7. If verified, the system **updates the borrow record**.
8. The database confirms success.
9. The system sends a **confirmation to the student**.

3. Use Case Diagram – Library Book Borrowing System



This diagram shows **what the system can do and who uses it.**

Actors:

- **Student**
- **Librarian**

Student Actions:

- Search Book
- View Book Details
- Borrow Book
- Return Book

Librarian Actions:

- Add Book
- Remove Book
- Update Book Info
- Manage Borrow Records